Economics of education

6th class

Information about the group presentation

- 3rd & 4th of March
- Randomize who presents in each class
- Everyone needs to submit their printed A3 poster in the beginning of the 3rd of March class

More info on group work

- You can now see your group and add your country of choice
- The group work will be a A3 poster, submitted in paper in the beginning of class 7 (**3rd of March**)
- Everyone needs to submit at the same time, irrespective of whether you present in class 7 or 8
- Presentation will be in power point (or equivalent programs), where you can zoom in in different sections of your poster
- Presentations will be **10min** per group

More info on group work

- If you use LLM, I shouldn’t be able to tell
- Deductions for obviously LLM writing
- References should be from reputable, reliable sources
- For every piece of information, you should have a reference
- It should be easy to cross-check information with the references
- Presentations are in person

More info on group work

What should you include in the poster?

1. **Budgeting** and **funding** of the education system

2. **System structure**: how does a student go through the education system, what choices and constraints does she face? (minimum years of compulsory schooling, tracking, affordability/accessibility)

3. **Student Performance**: Information on PISA test results or other relevant indicators of student achievement

4. Optional: Challenges, advantages, or interesting features or curiosities about the educational system.

More info on group work

- 10 minutes is short
- It’s okay to be selective in your presentation
- Ideally all the relevant information is on the poster, and the presentation is clear, logical, and structured
- Be selective and strategic about how you present the information
- Do not put super small letters in your poster

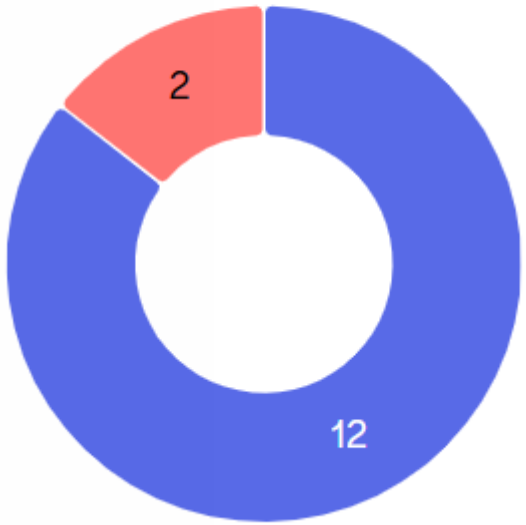
More info on group work

- Your group grade will be based on the content of your poster and on your presentation
- Your individual grade will be based on the group grade and peer evaluation

Questions?

Let’s go back to the last class

Consider the previous picture. Can we compare Ana and Felix earnings to infer the returns of schooling on earnings?



12	Yes since marginal slope is the same
2	No since Ana has higher returns to schooling
0	No since Ana and Felix have different schooling levels

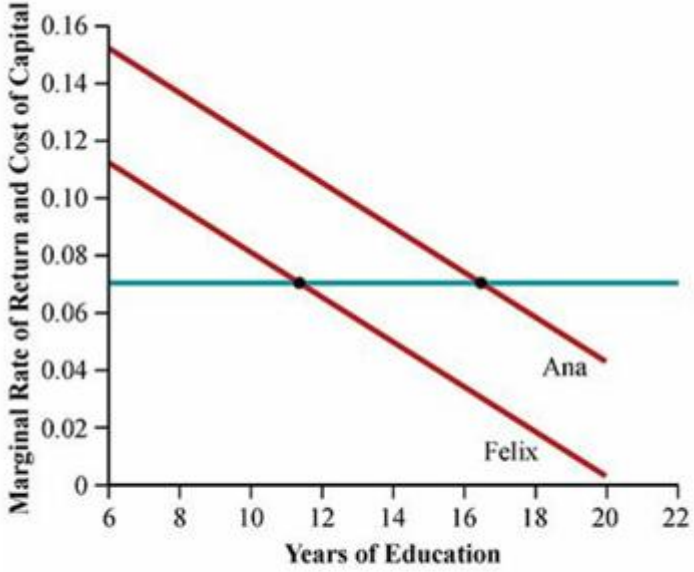


Figure 4.5
Lovenheim/Turner, *Economics of Education*, 1e,
© 2018 Worth Publishers

Can we compare Ana and Felix earnings?

Assume

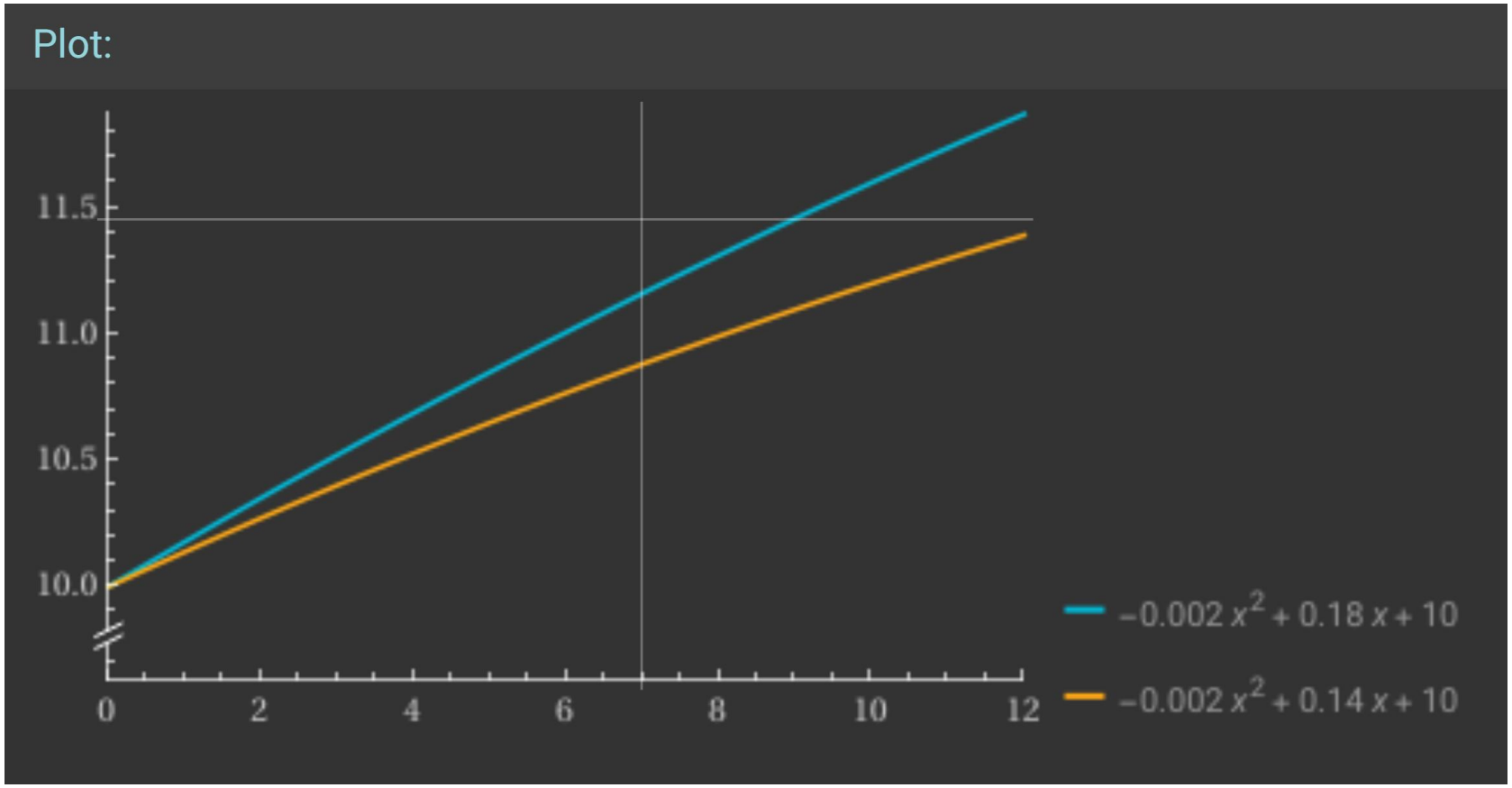
Ana:

$E(s) = 1 + 0.18s - 0.002s^2$; marginal returns $E'(s) = 0.18 - 0.004s$

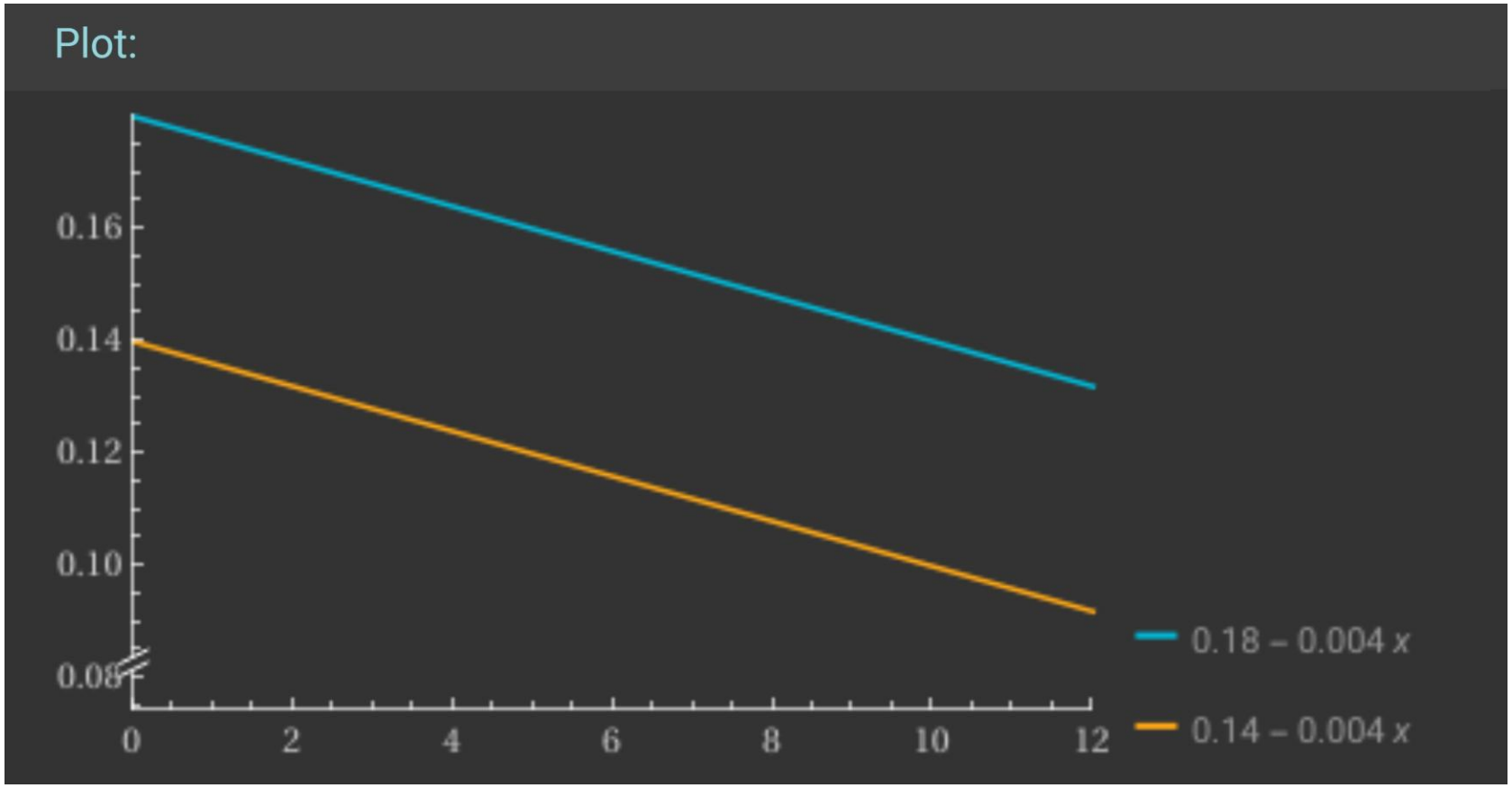
Felix:

$E(s) = 1 + 0.14s - 0.002s^2$; marginal returns$E'(s) = 0.14 - 0.004s$

Can we compare Ana and Felix earnings?



Can we compare Ana and Felix earnings?



Can we compare Ana and Felix earnings?

If we just compare outcomes; then

For Ana $E(17) = 1 + 0.18 * 17 - 0.002 * 17^2$ = 3.482

For Felix $E(12) = 1 + 0.14 * 12 - 0.002 * 12^2$ = 2.392

Earnings premium per school year:

$$(3.482 - 2.392)/\,(17 - 12) = 0.218$$

Can we compare Ana and Felix earnings?

The real return of education on earnings is:

$$p * (1 + 0.18s - 0.004s^2) + (1 - p)(1 + 0.14s - 0.004s^2)$$

Since there are only two individuals:

$p = 0.5$;

$$E(s) = 1 + 0.18s - 0.004s^2$$

$$E(17) - E(12)/5 = 2.904 - 2.584 / 5 = 0.32 / 5 = 0.064$$

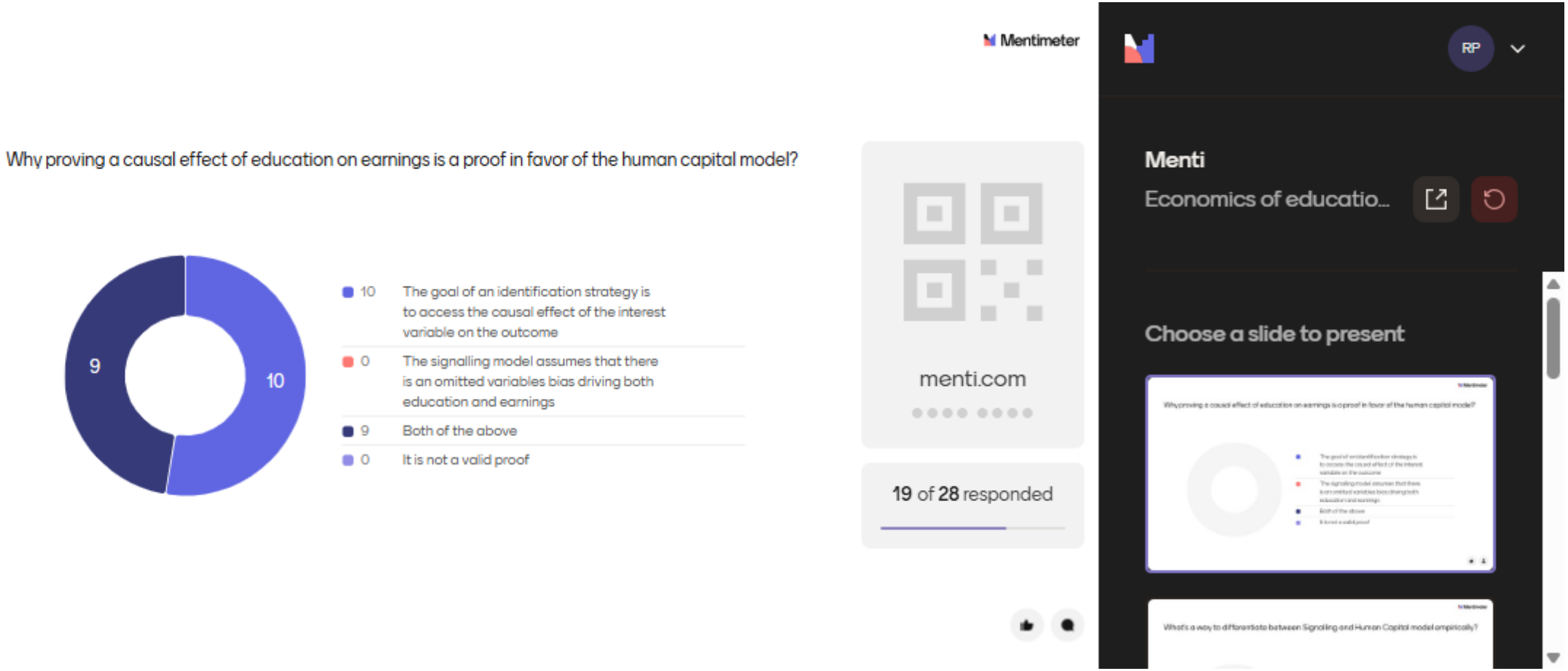
Can we compare Ana and Felix earnings?

True marginal average return of schooling 12 -17 years old: 0.064

Estimated marginal return for schooling 12 - 17 years old: 0.218

Complete miss!

Human capital versus signalling



Human Capital and Signalling model

Human capital theory: schooling increases human capital which increases productivity

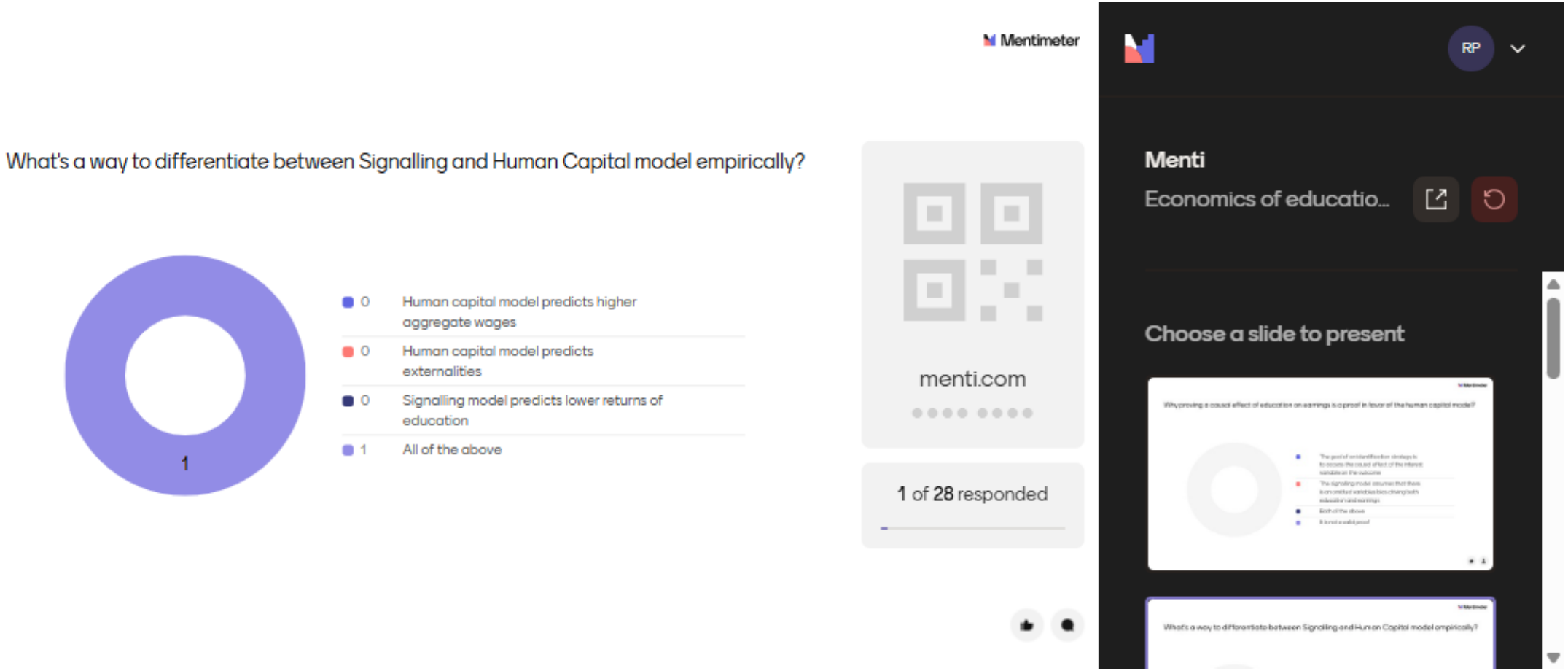
Signalling model: schooling is a way for high ability individuals to signal their ability

Human capital versus signalling

Consider the following equation:

$$E_i = \alpha + \beta s_i + \delta a_i + \varepsilon$$

Empirical tests



Empirical tests

- Both theories predict that schooling increases individual earnings
- How can we distinguish between the two empirically?

- Signalling model implies no positive **externalities** for education

Why?

Societal wise, it would be better for everyone not to get education and get an average wage (education is costly & there is no aggregate benefits in terms of wages)

Empirical tests

- In 3rd class we saw that there **is evidence for positive externalities** of education on total output/wages of lower educated
- Signalling model can’t explain this empirical regularity

Empirical tests

- Both theories predict that schooling increases individual earnings
- How can we distinguish between the two empirically?

- **Exogeneity in obtaining a degree**

Under human capital theory, having a diploma is irrelevant, only time in school that translates into human capital matters

Empirical tests

- US sample
- General Education Development (GED) credential is exploited as a signalling device
- GED is a set of exams that gives equivalence to a high-school diploma
- Paper focuses on doing the GED examination in 1990

Source: Tyler, J. H., Murnane, R. J., & Willett, J. B. (2000). Estimating the Labor Market Signaling Value of the GED. *The Quarterly Journal of Economics*, *115*(2), 431–468. https://doi.org/10.1162/003355300554818

Empirical tests

- There is a national minimum score, but states can set a higher minimum passing score
- Authors use a diff-in-diff
- Treatment group are those who have a grade just above the minimum score
- Control are individuals that had the same grade, but did not pass the threshold, as they faced a higher minimum passing score

Source: Tyler, J. H., Murnane, R. J., & Willett, J. B. (2000). Estimating the Labor Market Signaling Value of the GED. *The Quarterly Journal of Economics*, *115*(2), 431–468. https://doi.org/10.1162/003355300554818

Empirical tests

- Whites have a positive income effect of having a GED of 10-19%
- Minorities (Hispanic and black people) do not have the same positive effect
- Evidence for high school diploma as signalling device + discrimination across groups

TABLE V
DIFFERENCE-IN-DIFFERENCES ESTIMATES OF THE IMPACT OF THE GED ON 1995 EARNINGS OF DROPOUTS WHO TESTED IN 1990 (STANDARD ERRORS ARE IN PARENTHESES.)

	Experiment 4			Experiment 3			Experiment 3*		
	State passing standard is		Low-High standard contrast	State passing standard is		Low-High standard contrast	State passing standard is		Low-High standard contrast
	Low	High		Low	High		Low	High	
Panel A: Whites									
Test score is									
Low	9628	7849	1779	9362	7843	1509	9362	8616	746
	(361)	(565)	(670)	(400)	(312)	(507)	(400)	(219)	(456)
High	9981	9676	305	9143	9165	−23	9143	9304	−162
	(80)	(65)	(103)	(135)	(63)	(149)	(135)	(135)	(150)
Difference-in-differences for whites			1473*			1531**			907~
			(678)			(529)			(481)
Panel B: Minorities									
Test score is									
Low	6436	8687	−2252	7005	7367	−363	7005	6858	147
	(549)	(690)	(882)	(347)	(347)	(495)	(347)	(290)	(452)
High	7560	8454	−894	7782	8375	−593	7782	7568	214
	(184)	(96)	(207)	(214)	(93)	(233)	(214)	(133)	(252)
Difference-in-differences for minorities			−1357			231			−67
			(906)			(548)			(518)

Source: Tyler, J. H., Murnane, R. J., & Willett, J. B. (2000). Estimating the Labor Market Signaling Value of the GED. *The Quarterly Journal of Economics*, *115*(2), 431–468. https://doi.org/10.1162/003355300554818

Additional table info

** = significant at the 0.01 level, * = significant at the 0.05 level, ~ = significant at the 0.10 level.

Experiment 4: Test Score Low: score group = 4; Test Score High score groups = 5–10.

Passing Standard Low: 35 minimum score and 45 mean score; Passing Standard High: 40 minimum score and 45 mean score.

Low Passing Standard states: All states except for TX, LA, MS, NE, FL, NY, CA, WA, and CT; High Passing Standard states: NY and FL.

Experiment 3: Test Score Low: score group = 3; Test Score High score groups = 5–10.

Passing Standard Low: 40 minimum score *or* 45 mean score; Passing Standard High: 40 minimum score and 45 mean score.

Low Passing Standard states: TX, LA, MS, and NE; High Passing Standard states: NY and FL.

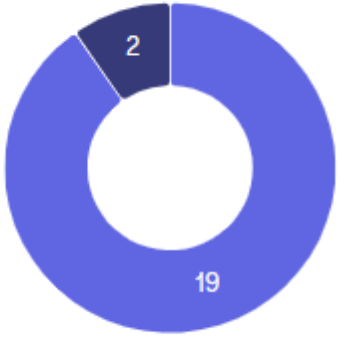
Experiment 3*: Test Score Low: score group = 3; Test Score High: score groups = 5–10.

Passing Standard Low: 40 minimum score *or* 45 mean score; Passing Standard High: 35 minimum score and 45 mean score.

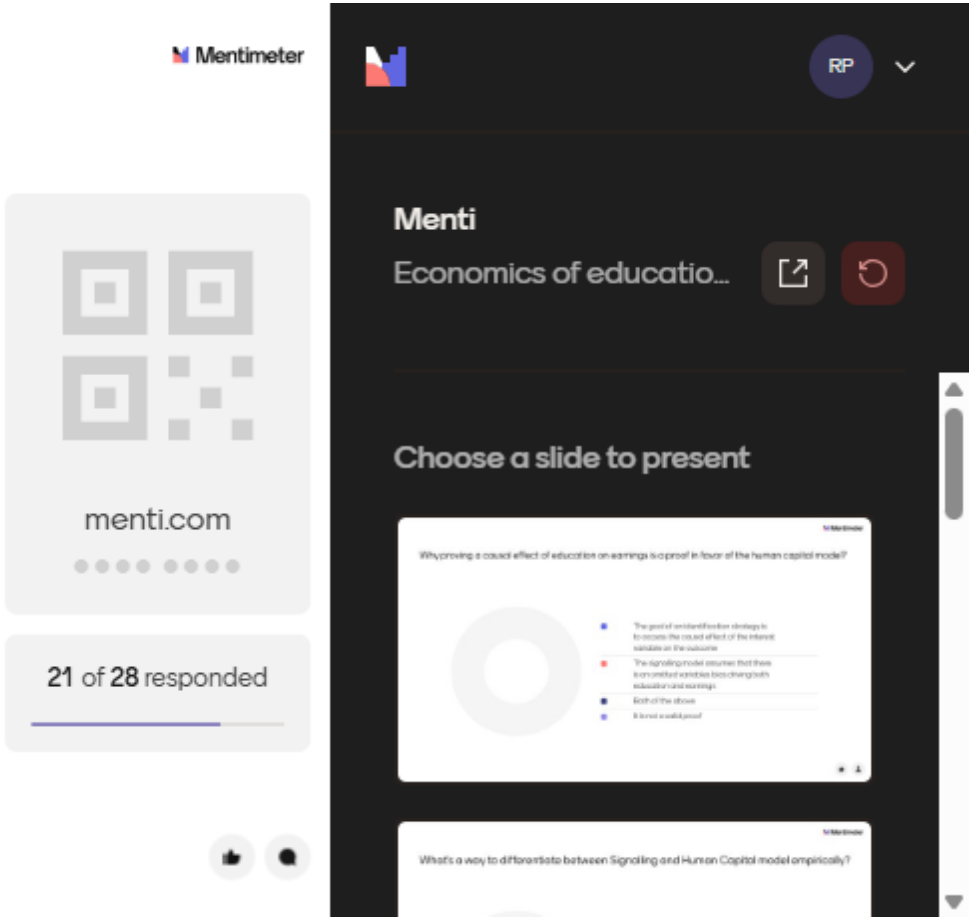
Low Passing Standard states: TX, LA, MS, and NE; High Passing Standard states: all states except TX, LA, MS, NE, NY, FL, and CT.

Quizz

According to this paper.....



19	White individuals signal is accepted in the labour market while minorities signal is not
0	White individuals signal is not accepted in the labour market whereas minorities signal is
2	White individuals earn more on average
0	There is no difference between white individuals and minorities when it comes to using a diploma as a signalling device



Empirical tests

- Texas sample
- High school exam in 1993–97
- Students are required to pass high school exit exams to have a high school diploma
- If they don’t pass, they need to re-take or repeat the year
- Not passing decreases your probability of graduating high school
- Fuzzy-RDD

Empirical tests

- First graph: no bunching!
- Second graph: jump in probability of graduating high school depending if you are above or below the threshold
- Human capital likely very similar around threshold

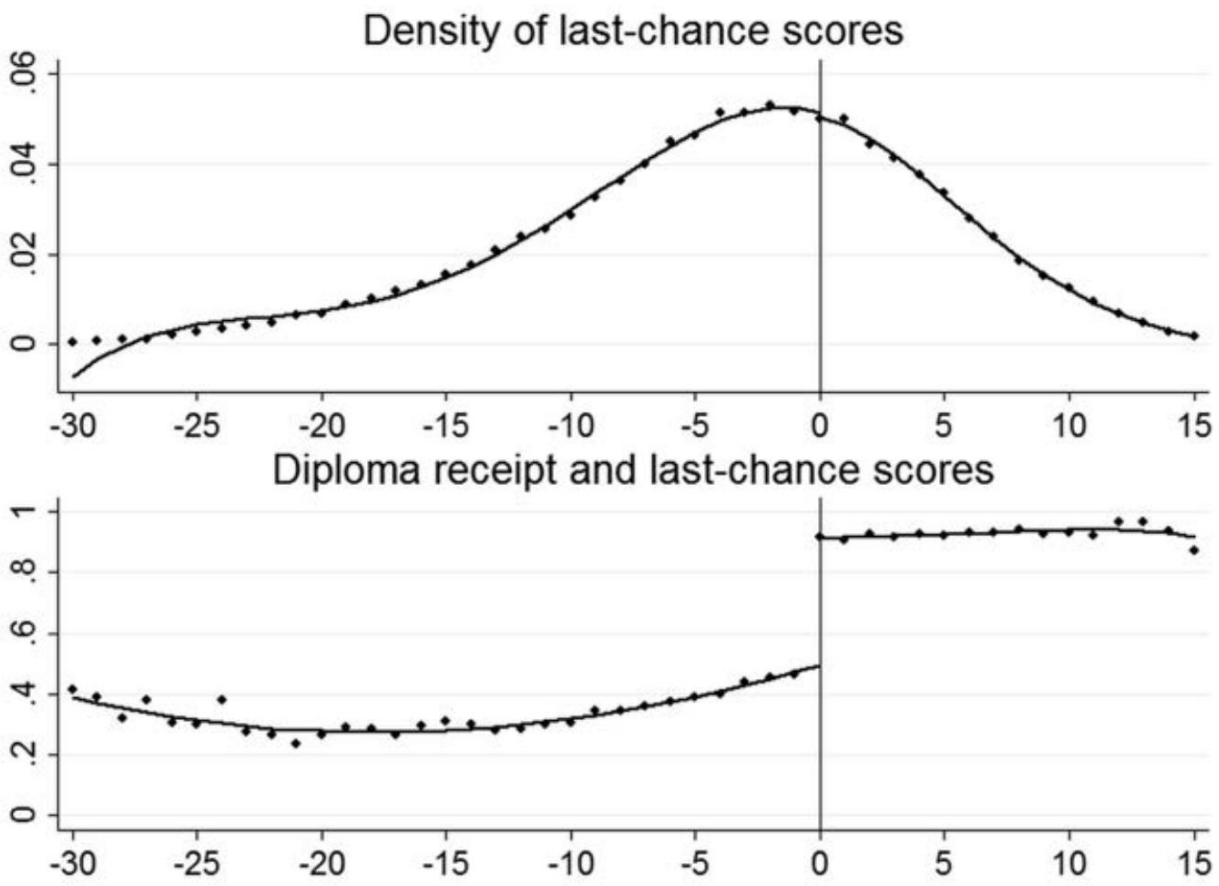


FIG. 1.—Last-chance exam scores and diploma receipt. The graphs are based on the last-chance sample. See table 1 and the text. Dots are test score cell means. The scores on the *x*-axis are the minimum of the section scores (recentered to be zero at the passing cutoff) that are taken in the last-chance exam. Lines are fourth-order polynomials fitted separately on either side of the passing threshold.

Source: Clark, D., & Martorell, P. (2014). The Signaling Value of a High School Diploma. *Journal of Political Economy*, *122*(2), 282–318. https://doi.org/10.1086/675238

Empirical tests

- No effect on income after 1-3, 4-6, 7-11, or 11 years
- High school diploma doesn't seem to play a signalling role here

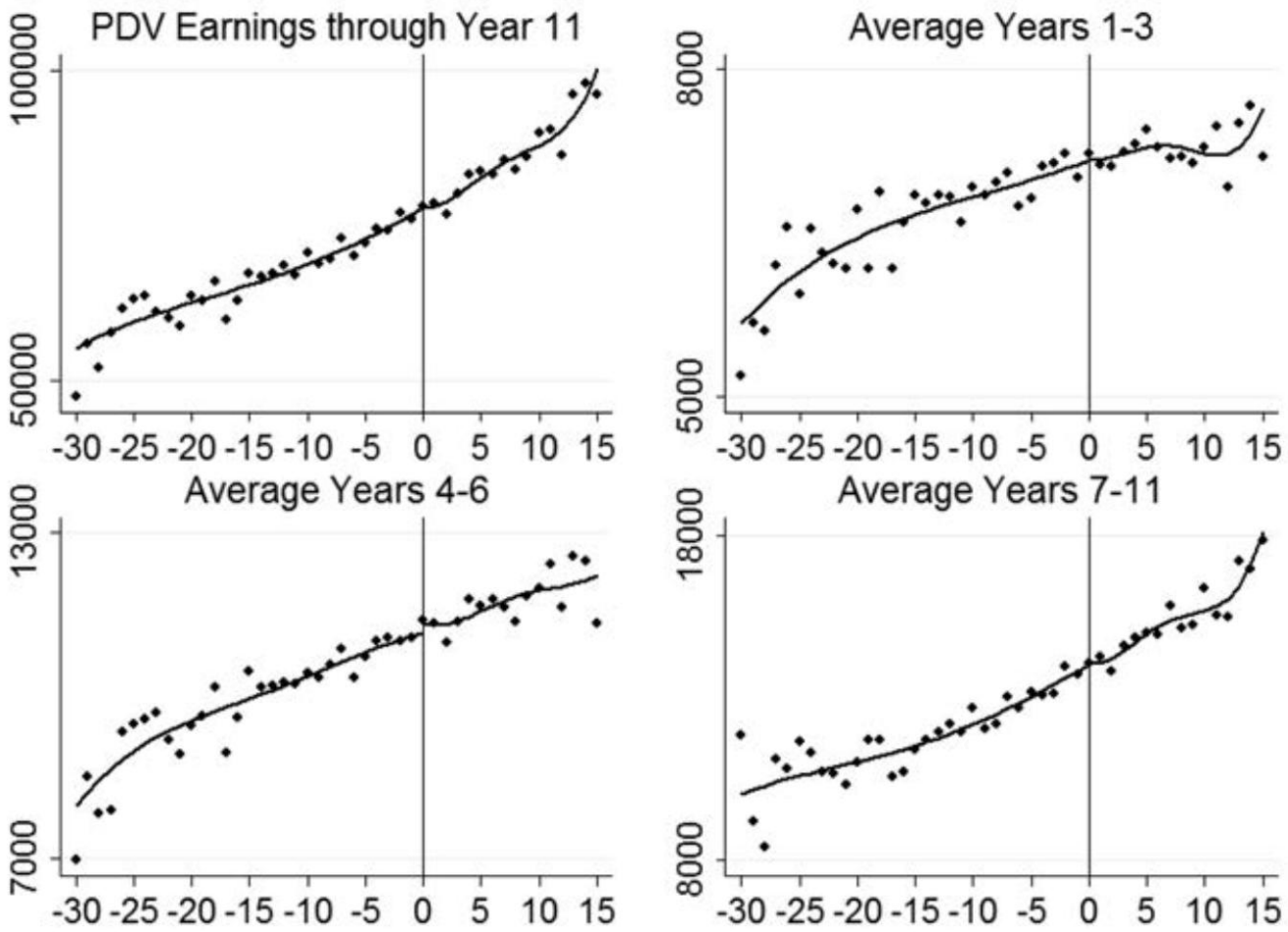


FIG. 2.—Earnings by last-chance exam scores. The graphs are based on the last-chance samples. See table 1 and the text. Dots are test score cell means. The scores on the *x*-axis are the minimum of the section scores (recentered to be zero at the passing cutoff) that are taken in the last-chance exam. Lines are fourth-order polynomials fitted separately on either side of the passing threshold.

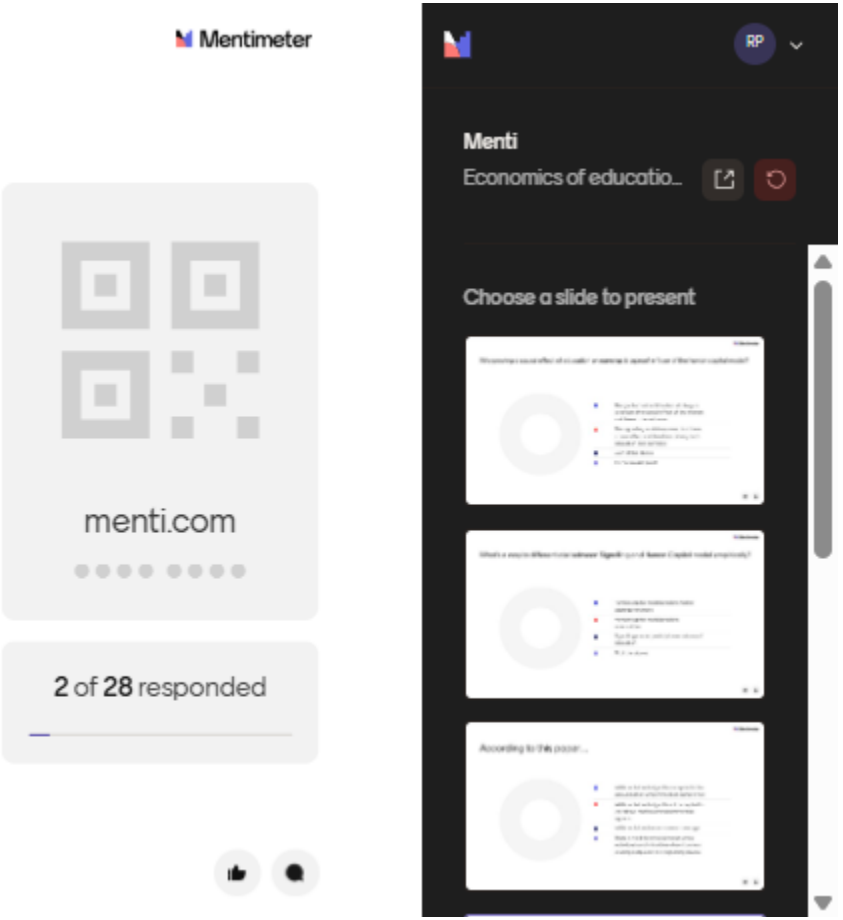
Source: Clark, D., & Martorell, P. (2014). The Signaling Value of a High School Diploma. *Journal of Political Economy*, *122*(2), 282–318. https://doi.org/10.1086/675238

Quizz

Note: identification strategy described in previous 3 slides

What is the reasoning behind this identification strategy?

- 0 People with and without diploma are equal on average
- 0 The treatment (=diploma) is randomly allocated
- 2 People with grades around the threshold are very similar to each other, and hence the only difference is the treatment (=diploma) ← Correct answer



Empirical tests

- Both theories predict that schooling increases individual earnings
- How can we distinguish between the two empirically?

- **Sheepskin effect:** the return to a year of education is higher when that year includes the awarding of a degree or education credential

Empirical tests

- US sample
- Compares the percentage change in earnings (Y axis) with years of schooling and diploma years (X axis)

Empirical tests

- Diploma years associate with higher earnings
- The jump in the return to education at year 16 is the only statistically significantly different from zero.
- This suggests a sizable **sheepskin effect** of graduating from college

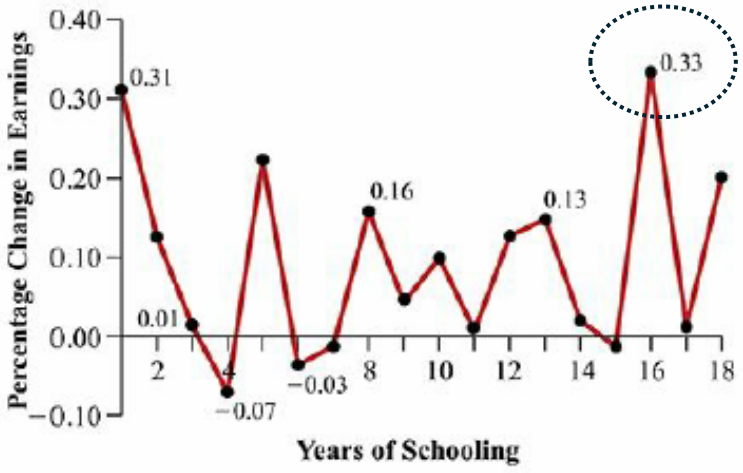


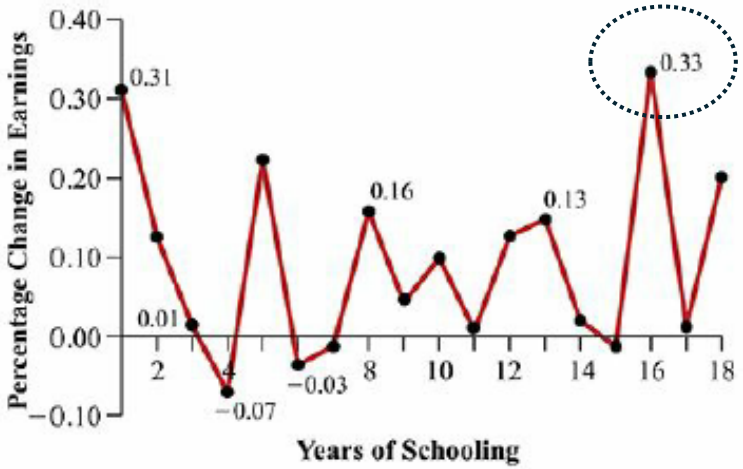
Figure 5.7
Lovenheim/Turner, *Economics of Education*, 1e
Data from: authors' calculations from Hungerford and Solon (1987), Table 2.

Figure 5.7 The Relationship Between Earnings and Schooling

There is an association between each incremental year of schooling and earnings changes in percent. For example, individuals obtaining 12 years of schooling earn 13% more than those obtaining 11 years of schooling.

Empirical tests

- What do you think about this analysis?



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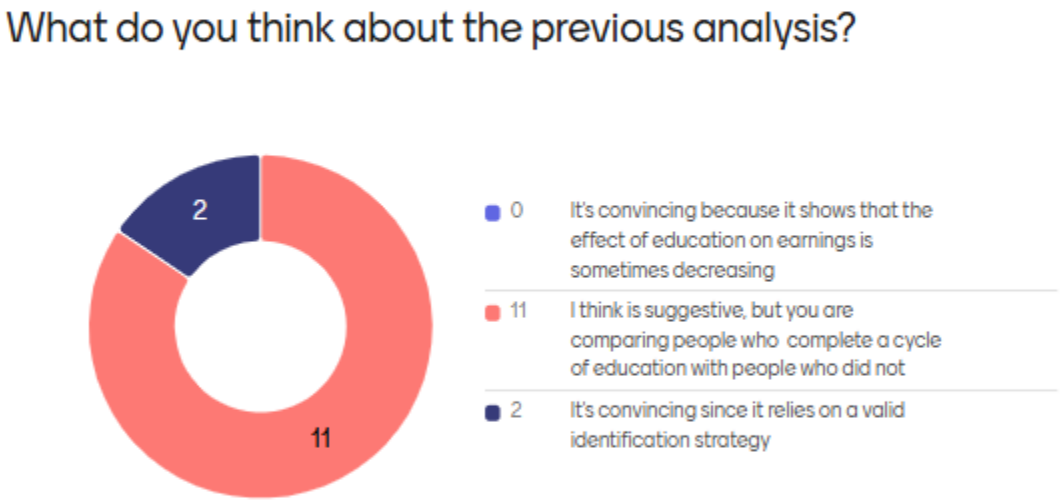
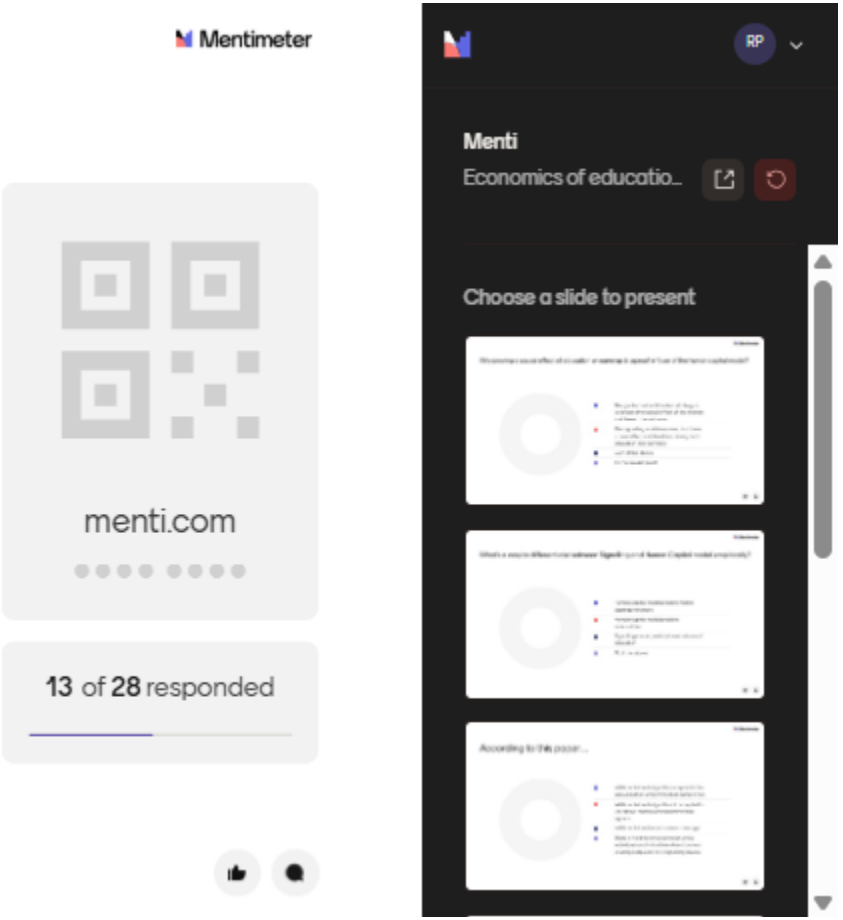
Quiz


What do you think about the previous analysis?
2
11
0 It's convincing because it shows that the effect of education on earnings is sometimes decreasing
11 I think is suggestive, but you are comparing people who complete a cycle of education with people who did not
2 It's convincing since it relies on a valid identification strategy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Economics of educatio...
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Empirical tests

- Dutch data set
- Different measures of years of education:

1. Actual years of education
2. Effective years of education (minimum necessary for degree)
3. Skipped years
4. Repeated years
5. Inefficient years (tracking + going from lower to higher levels)
6. Dropout years (years in education without diploma)

Empirical tests

Theoretical predictions under the signalling model:

- Skipping years increases earnings
- Repeating years decreases earnings
- Years spent in education without degree should not increase earnings

Empirical tests

Actual years of education

Effective years

Skipped years

Repeated years

Inefficient years

Dropout years

TABLE 1.—EARNINGS EQUATIONS: MALES

Variable	Equation (1)	Equation (2)	Equation (3a)	Equation (3b)
constant	1.339 (4.0)[a]	1.305 (3.8)[a]	1.295 (3.8)[a]	1.653 (5.2)[a]
s_a	0.043 (8.7)[a]			
s_e		0.050 (8.3)[a]	0.065 (9.1)[a]	0.071 (10.2)[a]
s_s			−0.089 (2.9)[a]	−0.093 (3.0)[a]
s_r			0.005 (0.2)	0.009 (0.5)
s_i			−0.014 (0.9)	−0.012 (0.8)
s_d			0.050 (4.3)[a]	0.056 (4.9)[a]
$iq/10$	0.040 (3.4)[a]	0.046 (3.9)[a]	0.037 (3.2)[a]	
experience	0.061 (2.2)[a]	0.063 (2.3)[a]	0.057 (2.1)[a]	0.056 (2.0)[a]
experience^2 / 10	−0.012 (2.0)[a]	−0.013 (2.2)[a]	−0.010 (1.7)	−0.010 (1.6)
Adjusted R^2	0.157	0.151	0.171	0.163
No. of cases	928	928	928	928

Note: t-values in parentheses.
[a] Significant at the 5% level.

Empirical tests

Actual years of education

Effective years

Skipped years

Repeated years

Inefficient years

Dropout years

TABLE 2.—EARNINGS EQUATIONS: FEMALES

Variable	Equation (1)	Equation (2)	Equation (3a)	Equation (3b)
constant	1.245 (2.0)[a]	1.504 (2.4)[a]	1.619 (2.6)[a]	1.886 (3.7)[a]
s_a	0.083 (4.9)[a]			
s_e		0.111 (5.6)[a]	0.123 (4.7)[a]	0.130 (5.4)[a]
s_s			−0.038 (0.4)	−0.032 (0.3)
s_r			−0.005 (0.1)	−0.005 (0.1)
s_i			0.014 (0.3)	0.012 (0.2)
s_d			0.093 (1.8)	0.098 (1.9)
$iq/10$	0.053 (1.3)	0.036 (0.9)	0.030 (0.0)	
experience	0.058 (1.0)	0.044 (0.8)	0.023 (0.4)	0.025 (0.4)
experience$^2/10$	−0.024 (1.7)	−0.020 (1.4)	−0.014 (1.0)	−0.014 (1.0)
Adjusted R^2	0.225	0.243	0.242	0.243
No. of cases	289	289	289	289

Note: t-values in parentheses.
[a] Significant at the 5% level.

Empirical tests

For men:

- Negative impact of skipping years
- Positive impact of dropout years
- Favours human capital model

For women:

- Few observation; likely little power to detect effects

Empirical tests

- Both theories predict that schooling increases individual earnings
- How can we distinguish between the two empirically?

- **Ripple effect:** after a schooling reform that increases school leaving age, educational attainment increases even for those unaffected by the reform

Empirical tests

- US Census data
- Compulsory Attendance Law (CAL)
- Older children (16/17) increase their enrolment if a new law is introduced that increases education of younger cohorts
- Consistent with signalling model

Source: Lang, K., & Kropp, D. (1986). Human capital versus sorting: the effects of compulsory attendance laws. *The Quarterly Journal of Economics*, *101*(3), 609-624.

TABLE II

DETERMINANTS OF THE LOG ODDS ENROLLMENT RATE

	16-Year-olds	17-Year-olds
CAL12/14	0.126	0.028
	(0.124)	(0.115)
CAL15	0.272	0.168
	(0.149)	(0.139)
CAL16	0.258	0.143
	(0.123)	(0.115)
CAL17/18	0.497	–
	(0.139)	
CAL17	–	0.335
		(0.131)
CAL18	–	0.308
		(0.131)
1920	−0.093	−0.078
	(0.097)	(0.091)
1930	0.423	0.399
	(0.102)	(0.095)
1950	1.144	1.183
	(0.105)	(0.098)
1960	1.587	1.660
	(0.105)	(0.098)
1970	2.113	2.254
	(0.106)	(0.099)
Constant	0.041	−0.553
	(0.103)	(0.096)

Standard errors are in parentheses.

ROSLA directly impacts CSE

Empirical tests

- Uk data set
- Effect of ROSLA on higher education qualifications
- No effect for men or women above CSE after controlling for paternal SES
- Note: t-stat > 2 for significance

Table 10

Effect of Minimum School Leaving Age on the Probability of Achieving Qualification Levels

	CSE	O-level	A-level|O-level	Degree|A-level
Men				
(1)	0.101 (4.04)	0.022 (0.78)	0.056 (1.44)	−0.098 (1.80)
(2) = (1) + paternal SES	0.105 (4.24)	0.024 (0.83)	0.057 (1.44)	−0.097 (1.78)
$\chi^2(6)$*	356.5	530.4	93.5	30.5
(3) = (2) + SES*SLA	0.070 (1.74)	−0.022 (0.52)	0.068 (1.33)	−0.115 (1.67)
$\chi^2(6)$*	233.1	275.2	25.1	20.1
$\chi^2(6)$†	4.8	3.5	7.9	6.8
Observations	5,166	5,166	2,650	1,268
Women				
(1)	0.075 (2.59)	0.078 (2.10)	0.018 (0.38)	−0.040 (0.54)
(2) = (1) + paternal SES	0.070 (1.74)	−0.022 (0.52)	0.068 (1.33)	−0.115 (1.67)
$\chi^2(6)$*	181.0	330.4	93.5	30.5
(3) = (2) + SES*SLA	0.034 (0.77)	−0.017 (0.31)	0.028 (0.44)	−0.020 (0.22)
$\chi^2(6)$*	106.7	154.1	25.1	20.10
$\chi^2(6)$†	3.5	7.03	7.9	6.8
Observations	2,812	2,812	1,836	768

Note: General Household Surveys 1982–92 (odd years), cohort born 1953–63. Robust t-statistics in parentheses.
Model 1 also includes dummies for birth year, parental origin, survey years and smoking behaviour at 16.
*F-test for joint significance of the paternal Socio Economic Status dummies. The critical value for χ^2 (6) = 12.6 at the 5% level.
†F-test for joint significance of the interactions between paternal Socio Economic Status dummies and minimum school leaving age.
The critical value for $\chi^2(6)$ = 12.6 at the 5% level.

Source: Chevalier, A., Harmon, C., Walker, I., & Zhu, Y. (2004). Does education raise productivity, or just reflect it?. *The Economic Journal*, *114*(499), F499-F517.

Empirical tests

- No evidence for "ripple" effect here, supporting Human Capital hypothesis

Table 10

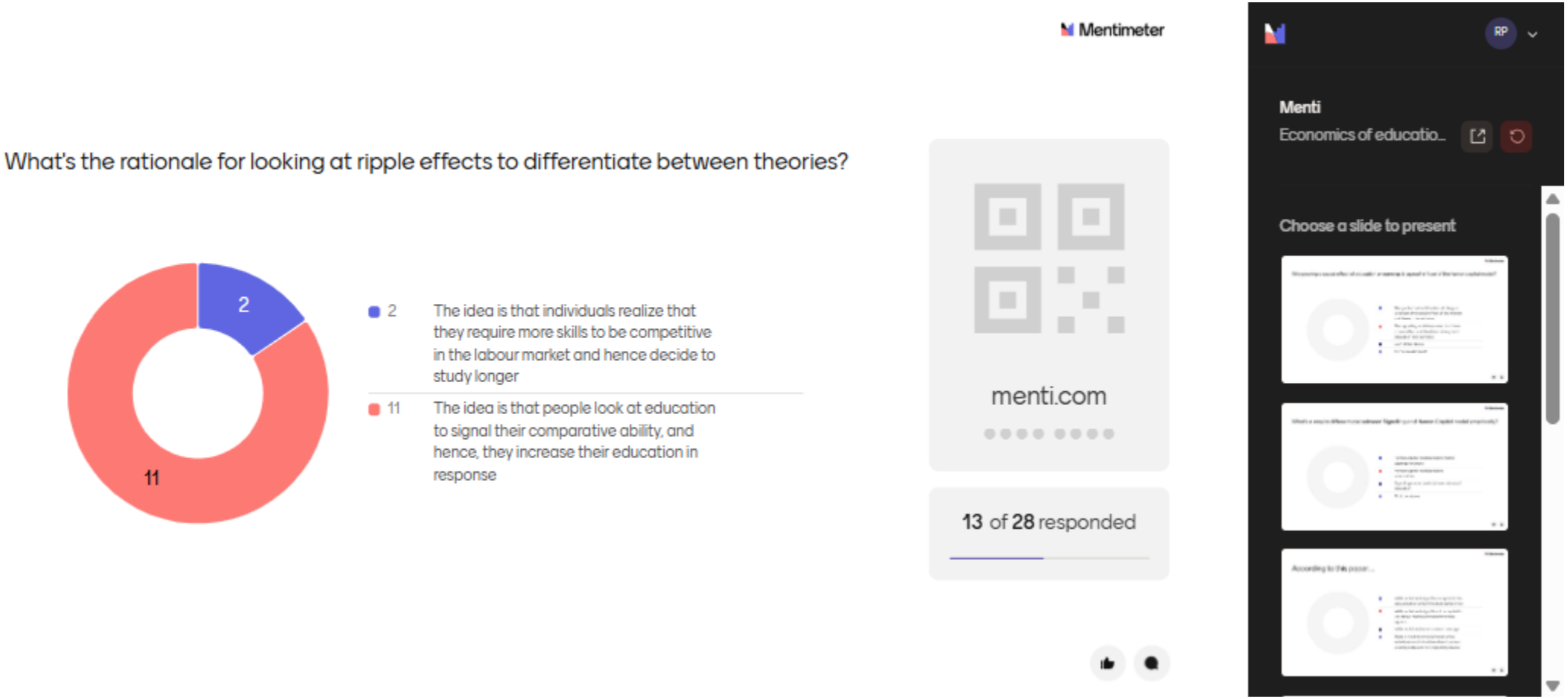
Effect of Minimum School Leaving Age on the Probability of Achieving Qualification Levels

	CSE	O-level	A-level|O-level	Degree|A-level
Men				
(1)	0.101 (4.04)	0.022 (0.78)	0.056 (1.44)	−0.098 (1.80)
(2) = (1) + paternal SES	0.105 (4.24)	0.024 (0.83)	0.057 (1.44)	−0.097 (1.78)
$\chi^2(6)$*	356.5	530.4	93.5	30.5
(3) = (2) + SES*SLA	0.070 (1.74)	−0.022 (0.52)	0.068 (1.33)	−0.115 (1.67)
$\chi^2(6)$*	233.1	275.2	25.1	20.1
$\chi^2(6)$†	4.8	3.5	7.9	6.8
Observations	5,166	5,166	2,650	1,268
Women				
(1)	0.075 (2.59)	0.078 (2.10)	0.018 (0.38)	−0.040 (0.54)
(2) = (1) + paternal SES	0.070 (1.74)	−0.022 (0.52)	0.068 (1.33)	−0.115 (1.67)
$\chi^2(6)$*	181.0	330.4	93.5	30.5
(3) = (2) + SES*SLA	0.034 (0.77)	−0.017 (0.31)	0.028 (0.44)	−0.020 (0.22)
$\chi^2(6)$*	106.7	154.1	25.1	20.10
$\chi^2(6)$†	3.5	7.03	7.9	6.8
Observations	2,812	2,812	1,836	768

Note: General Household Surveys 1982–92 (odd years), cohort born 1953–63. Robust t-statistics in parentheses.
Model 1 also includes dummies for birth year, parental origin, survey years and smoking behaviour at 16.
*F-test for joint significance of the paternal Socio Economic Status dummies. The critical value for χ^2 (6) = 12.6 at the 5% level.
†F-test for joint significance of the interactions between paternal Socio Economic Status dummies and minimum school leaving age.
The critical value for $\chi^2(6)$ = 12.6 at the 5% level.

Source: Chevalier, A., Harmon, C., Walker, I., & Zhu, Y. (2004). Does education raise productivity, or just reflect it?. *The Economic Journal*, *114*(499), F499-F517.

Quizz

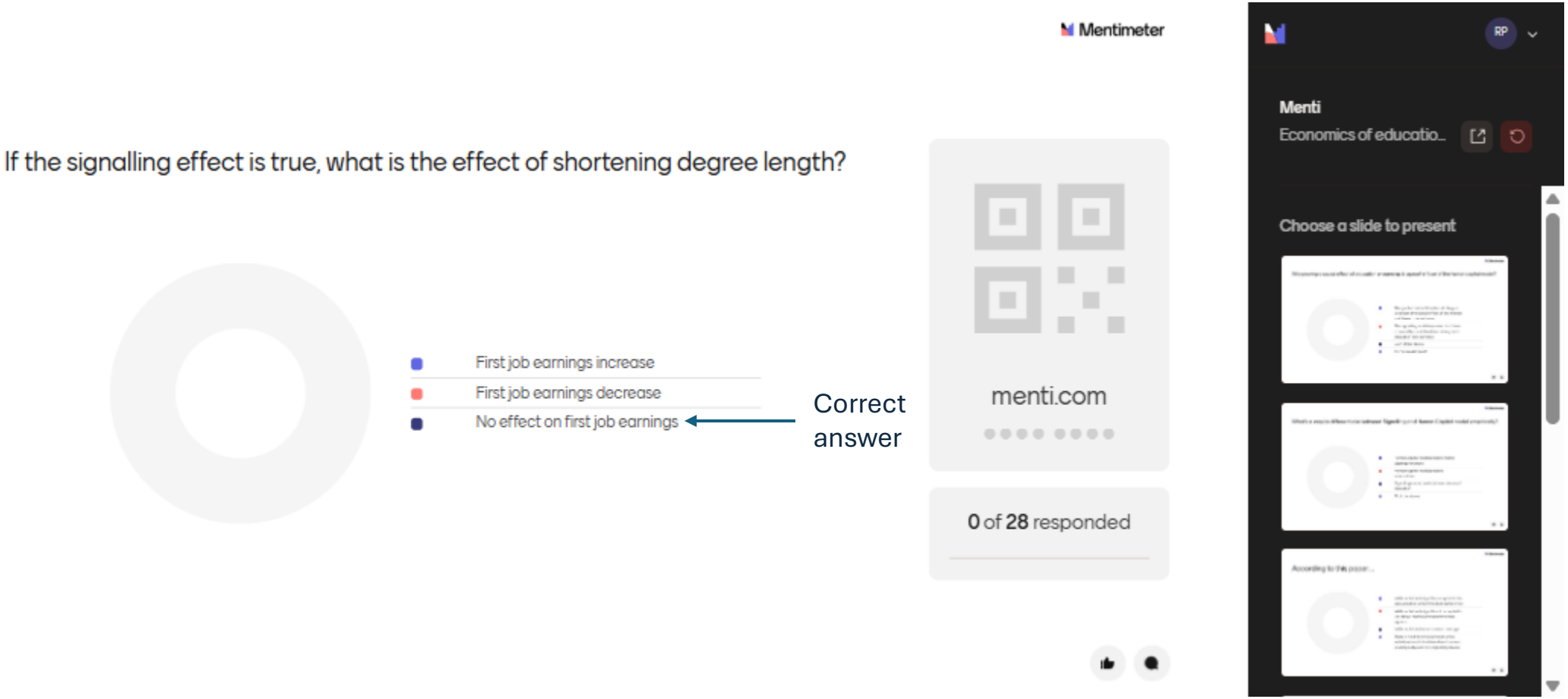

Mentimeter
What's the rationale for looking at ripple effects to differentiate between theories?
2
11
2 The idea is that individuals realize that they require more skills to be competitive in the labour market and hence decide to study longer
11 The idea is that people look at education to signal their comparative ability, and hence, they increase their education in response
menti.com
13 of 28 responded
RP
Menti
Economics of educatio...
Choose a slide to present

Empirical tests

- Both theories predict that schooling increases individual earnings
- How can we distinguish between the two empirically?

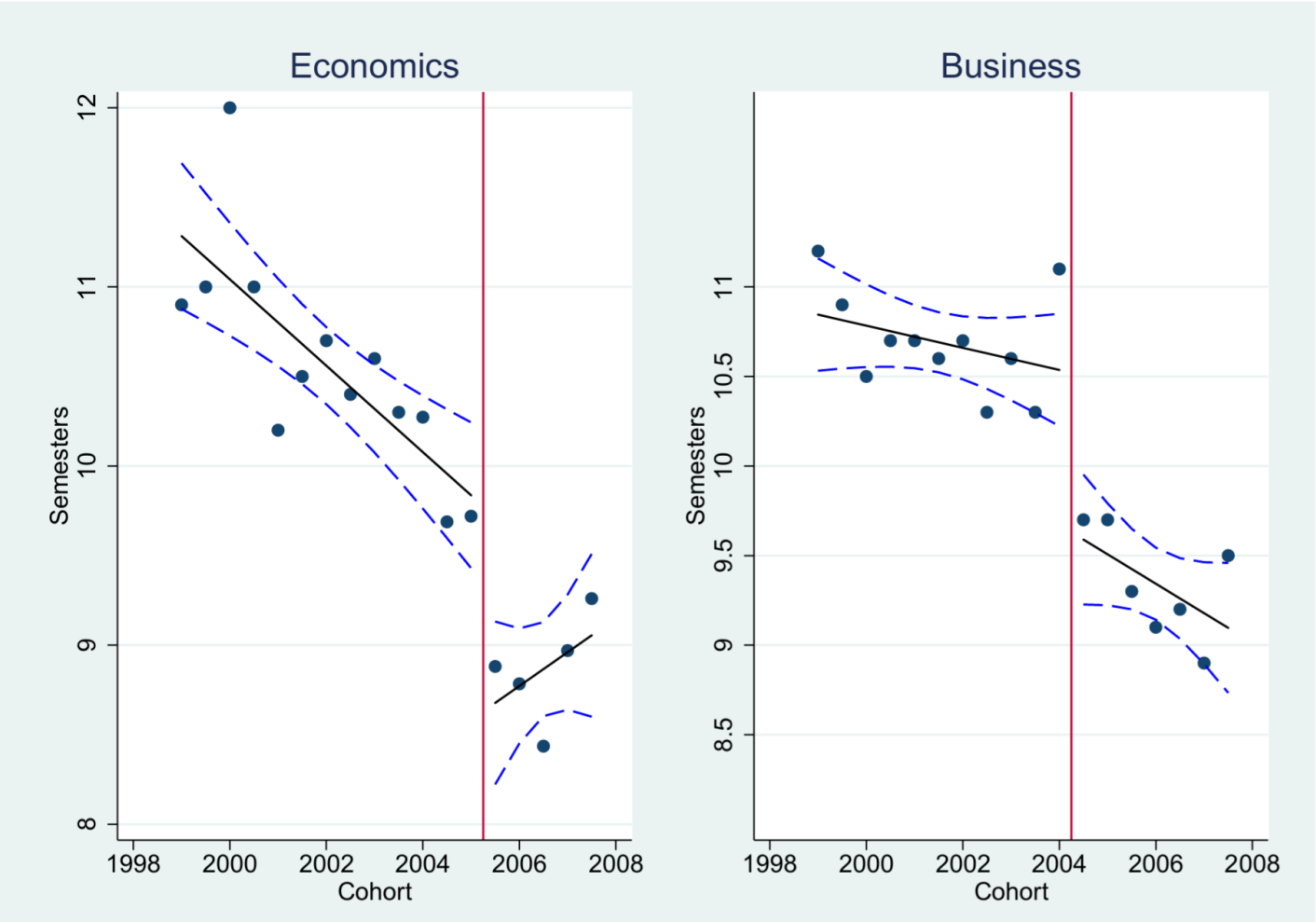
- **Effect of shortening college courses duration**

Quizz


Mentimeter
If the signalling effect is true, what is the effect of shortening degree length?
First job earnings increase
First job earnings decrease
No effect on first job earnings
Correct answer
menti.com
0 of 28 responded
RP
Menti
Economics of educatio...
Choose a slide to present

Empirical tests

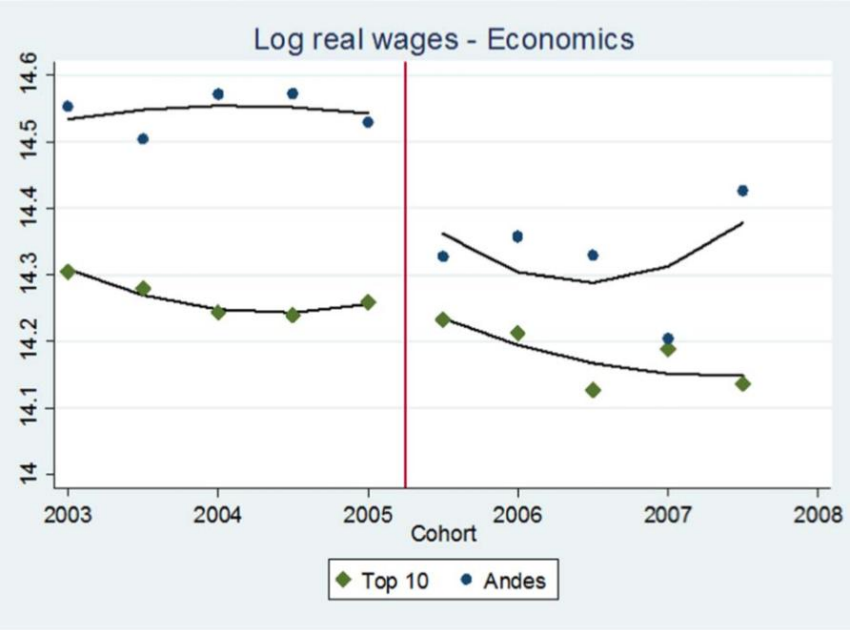
- Sample from a Colombia's university (Los Andes)
- Reform that decreased degree duration & credits necessary to complete degree
- No change in size of the entering class, their average high school exit exam scores, or graduation rates

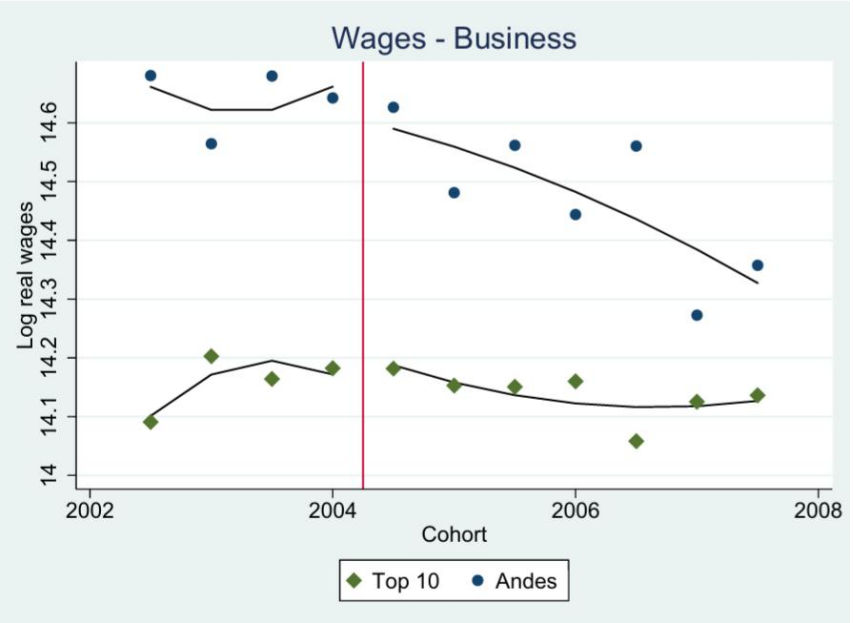


Source: Arteaga, C. (2018). The effect of human capital on earnings: Evidence from a reform at Colombia's top university. *Journal of Public Economics*, *157*, 212-225.

Empirical tests

- Figure compares top 10 Colombian Universities with Los Andes Uni
- Reduction of degree length reduced wages

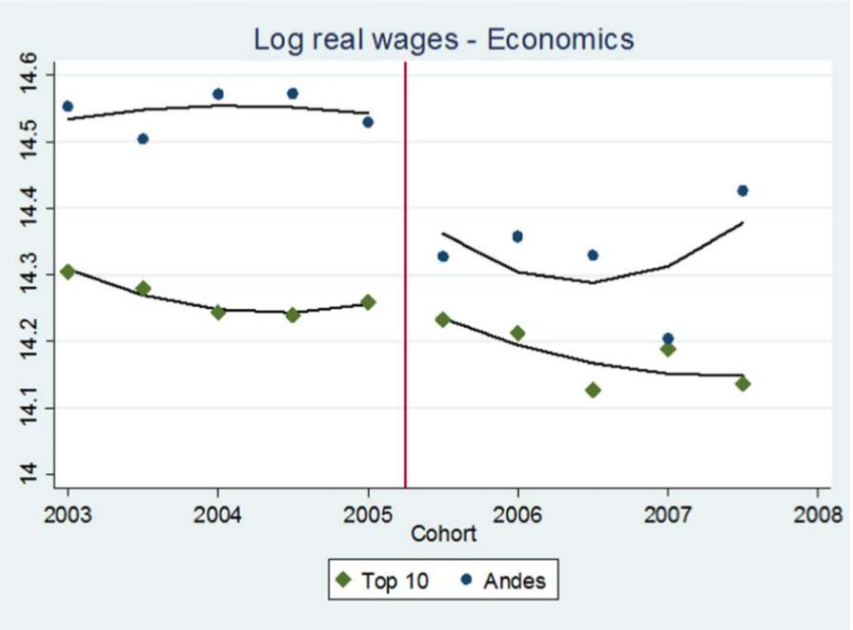


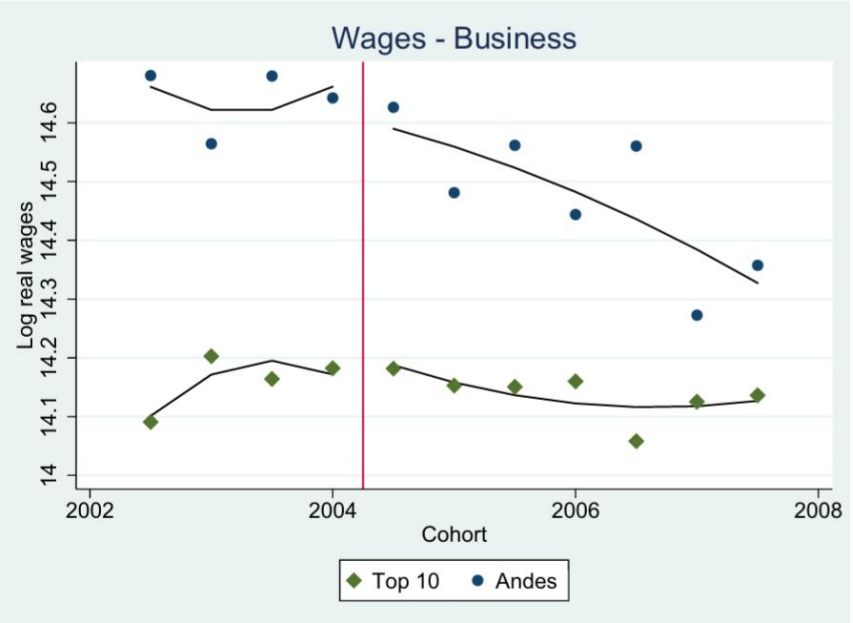


Source: Arteaga, C. (2018). The effect of human capital on earnings: Evidence from a reform at Colombia's top university. *Journal of Public Economics*, *157*, 212-225.

Empirical tests

- Through employer surveys, author finds that the reduction in wages may resulted from a decline in performance during the recruitment process
- Reform reduced probability of being hired by 17%
- Support for the human capital hypothesis





Source: Arteaga, C. (2018). The effect of human capital on earnings: Evidence from a reform at Colombia's top university. *Journal of Public Economics*, *157*, 212-225.

How to distinguish between the two

- Estimating causal effects of education
- Education externalities
- Exogenous effect of obtaining a degree
- Sheepskin Effect
- Ripple effect
- Varying degree length

Take-aways

- There is strong evidence for human capital model
- Nonetheless, in some contexts, signalling role of education matters
- Probably both are relevant
- Education must be seen as a mix between human capital (+++) and signalling device (+)

Questions?